

IonShield — Science Backgrounder

How measured space weather becomes a mission decision. One page, every model cited; every output labeled measured / modeled / doctrine.

The physical problem

The ionosphere (60–1,000 km altitude) is plasma created by solar radiation. Every RF signal that crosses it (GPS, SATCOM) or refracts off it (HF skywave) is affected by its free-electron content. Solar flares, coronal mass ejections, and energetic-particle events change that content on timescales of minutes (flares: 8-minute light-travel warning) to days (CMEs: 1–2 day transit). Four physically distinct effects follow; IonShield models each separately, per location, because geomagnetic latitude, local solar time, and equipment class control who gets hit.

Measured inputs (5-minute ingest cadence, SHA-256 provenance on every value)

Kp index — global geomagnetic disturbance, 0–9 (NOAA/GFZ magnetometers)

GOES X-ray flux (1–8 Å) — flare class A/B/C/M/X is log-scale flux

GloTEC — measured global Total Electron Content map

IMF B_z — solar-wind field N–S component, DSCOVR @ L1. Sustained southward (< -10 nT) reconnects with the geomagnetic field: the storm trigger

GOES ≥ 10 MeV protons — energetic particles funneled into the polar caps

Solar wind speed/density, F10.7, NOAA 3-day Kp forecast + forecaster R/S/G probabilities

Four effect models

GPS error — ionospheric group delay [Klobuchar 1996; Mannucci 2005 (storm enhancement)]

Free electrons delay the ranging code: ≈ 0.162 m of error per TECU at L1, $\times$ obliquity factor 2.5 (slant geometry), $\times$ storm scaling. Measured GloTEC used where available. Delay scales as $1/f^2$, so dual-frequency receivers (L1/L2, L1/L5) solve for and cancel it — single-frequency receivers (DAGR, Group 1 UAS) take the full error. That receiver-class split is why doctrine records 100 m horizontal / 200 m vertical on exactly that equipment.

HF blackout — D-layer absorption [CCIR-888 / Spencer 1971 (SID); Rose & Ziauddin 1962 (storm); Bailey 1964 (PCA)]

HF works by refracting off the F-layer; it dies when the lower D-layer absorbs it first. Three additive sources (dB): flare X-rays ionize the sunlit D-layer in minutes (solar-zenith-angle scaled); Kp-driven precipitation at high geomagnetic latitude; and polar cap absorption — solar protons (≥ 10 pfu, NOAA S1) blanketing HF poleward of $\sim 65^\circ$ for days. Total absorption vs ~ 25 dB fade margin $\rightarrow$ blackout probability. SINCGARS FM (30–88 MHz line-of-sight) never transits the ionosphere — immune; EHF passes through at f high enough that $1/f^2$ effects are negligible.

SATCOM dropout — scintillation [Basu 1988 / Aquino 2005 (S4 regimes); ITU-R P.531 Nakagami-m fading]

Storms make the ionosphere patchy; signals self-interfere and received power flickers — the S4 index. S4 estimated per regime (equatorial post-sunset plasma bubbles; Kp-driven auroral; proton-driven polar), then Nakagami-m statistics ($m \approx 1/S4^2$) convert S4 to fade depth and outage probability against the link margin. The physics behind UHF TACSAT 'broken and unreadable.'

Radar degradation — range bias + clutter [Skolnik (group delay $40.3 \cdot \text{TEC}/f^2$); CCIR Faraday rotation]

Radar ranges by echo timing; ionospheric group delay adds $40.3 \cdot \text{TEC}/f^2$ metres of apparent range plus polarization rotation. Scales as $1/f^2$: L-band counter-battery radar (AN/TPQ-53) hit hardest, X-band barely affected — plus auroral clutter producing false/missed tracks.

From physics to verdict

Per-waypoint metrics (GPS error in metres, HF absorption in dB, S4, fade probability) are computed at each location — geomagnetic latitude, dayside/nightside, polar cap membership all matter. A mission layer applies the operator's tolerances (0.5 m breaks RTK survey; 25 m is fine for a patrol) to produce CLEAR / CAUTION / HIGH RISK / DELAY. On top, a doctrine rule library (5 equipment classes $\times$ 3 weather states, each rule citing ALSSA 'True Impacts of Space Weather on a Ground Force' or NOAA scale definitions) speaks in equipment names — the physics carries the numbers underneath. ML (ridge-regression Kp forecaster, event classifier) forecasts drivers only; it never overrides the physics. Recommendations are probabilistic windows, never binary go/no-go.

Honest limitations: propagation models are published simplified analytical forms, not ray-tracing or assimilative ionosphere models. S4 lacks solar-cycle/seasonal terms; HF output is not yet frequency-specific; GPS error awaits validation against CORS receiver data during storms (planned Phase II with USSF weather officers). Every output is labeled measured / modeled / doctrine-derived, and historical replay (e.g. May 2024 Gannon G5 storm, GFZ definitive Kp) lets the engine be checked against documented events.

IonShield · operational space weather intelligence · live NOAA SWPC ingest · UDS/Zarf air-gap deployable · ATAK KML/CoT outputs